

EarningsEdge AI

Earnings Call Sentiment & Alpha Signal Generator

Project Proposal | BS-AI Capstone | March 2026

One-Line Pitch

EarningsEdge is an NLP-powered platform that extracts management sentiment, hedging language, and forward-guidance signals from earnings call transcripts — and translates them into actionable alpha signals correlated with post-call price movement.

1. Market Need

The buy-side research process is drowning in unstructured text. Every quarter, publicly traded U.S. companies hold over 5,000 earnings calls. Each transcript runs 8,000–15,000 words and contains critical non-financial signals: management tone shifts, increased hedging language, forward-guidance ambiguity, and analyst pushback patterns. These signals have documented predictive power — yet they are largely missed at scale.

Metric	Value
Earnings calls per quarter (U.S. public companies)	~5,000+
Average transcript length	8,000 – 15,000 words
Avg. analyst time per call transcript	45 – 90 minutes
Calls a senior analyst covers per quarter	40 – 80
Hours spent on transcript analysis per quarter	~60 – 120 hrs
U.S. buy-side AUM subject to earnings signals	\$30+ trillion

2. User Pain Points

Applying the 'observe the professional's day' framework to a buy-side equity analyst reveals the following friction points:

2.1 The Analyst's Workflow (Morning of Earnings Day)

- 6:30 AM — Earnings release drops; analyst downloads transcript PDF from IR site or Bloomberg
- 7:00 AM — Manually skims 10,000-word document looking for guidance language changes
- 7:30 AM — Compares CEO tone against prior quarter transcript from memory or notes
- 8:00 AM — Joins live call, tries to annotate tone while listening
- 9:00 AM — Writes analyst note summarizing sentiment — largely subjective
- Post-market — Reviews 3–5 more transcripts from the same day; backlog grows

2.2 The Core Pain Points

- **Volume problem:** 500+ calls in a 3-week earnings window; no analyst team can keep up.
- **Subjectivity problem:** Two analysts read the same transcript and reach different tone assessments.
- **Speed problem:** Institutional desks often trade within minutes of call end; manual analysis is too slow.
- **Recall problem:** QoQ tone comparison requires remembering or digging up prior call notes.
- **Signal gap:** Linguistic features like hedge word density, certainty language, and CFO-vs-CEO divergence are never systematically extracted.

3. Proposed Solution

EarningsEdge is a three-layer NLP pipeline that converts raw transcript text into structured sentiment scores and tradeable alpha signals.

3.1 Data Ingestion Layer

- Source transcripts via SEC EDGAR full-text search (free), MotleyFool/Seeking Alpha scraper, or Hugging Face datasets (EarningsCall dataset — 2,700+ transcripts)
- Price data ingested via Alpaca Markets API (free, real-time and historical OHLCV)
- Pre-processing: speaker diarization (CEO vs. CFO vs. Analyst Q&A separation), sentence segmentation, domain tokenization

3.2 NLP Model Layer (Recommended: FinBERT + LLM Augmentation)

Model Recommendation: FinBERT fine-tuned on earnings call transcripts

FinBERT (ProsusAI) is pre-trained on 1.8M financial news sentences and outperforms general BERT on finance sentiment by ~15% F1. For this MVP, we fine-tune FinBERT on the EarningsCall dataset for transcript-specific sentiment, then augment with a prompted LLM (GPT-4o or Claude) for qualitative hedging-language extraction — extracting features that classification models miss.

- Sentiment scoring: Fine-tuned FinBERT classifies each sentence as positive/neutral/negative
- Hedge detection: Custom NER model identifies hedge phrases ('we expect', 'subject to', 'if conditions permit') using a compiled hedge lexicon (Loughran-McDonald wordlist)
- Management confidence score: Composite of certainty language ratio, forward-guidance specificity, and tone delta vs. prior quarter
- Analyst pushback signal: Sentiment of Q&A portion classified separately from prepared remarks — divergence is a key signal

3.3 Alpha Signal Layer

- Cross-reference sentiment scores with 1-day, 3-day, and 5-day post-call price returns (Alpaca API)
- Backtest: Train signal model on 2019–2023 transcripts, test on 2024 out-of-sample
- Output: Per-call composite score (-1 to +1) with historical Sharpe attribution
- Deliverable: REST API endpoint returning JSON signal per ticker + call date

4. How We Measure Success

Metric	Category	Target (MVP)	Stretch Goal
F1 Score (Sentiment Model)	NLP Quality	> 0.78	> 0.85
Hedge Phrase Recall	NLP Quality	> 0.80	> 0.88
Signal-Return Correlation (Pearson r)	Alpha Quality	> 0.20	> 0.35
Backtest Sharpe Ratio (long/short)	Alpha Quality	> 0.8	> 1.2
API Latency (per transcript)	System	< 5 seconds	< 2 seconds
Transcripts processed (backtest)	Coverage	500+	2,000+
False positive rate (signals)	Reliability	< 20%	< 12%

5. Data Sources

Source	Use Case
SEC EDGAR Full-Text Search API (free)	Official earnings call transcripts (8-K filings)
Hugging Face: EarningsCall Dataset	Pre-labeled transcripts for fine-tuning FinBERT
Alpaca Markets API (free tier)	Historical + live OHLCV price data for signal backtesting
Loughran-McDonald Financial Wordlist	Hedge phrase and uncertainty lexicon for NER features
FinNLP (GitHub)	Additional financial text datasets for domain adaptation
WRDS / Compustat (if accessible)	Fundamental data to control for earnings surprise vs. tone signal

6. Feasibility Analysis

6.1 Technical Feasibility

- FinBERT fine-tuning on 2,700 transcripts: ~4–6 hours on a single A100 GPU (Google Colab Pro, ~\$50)
- MVP pipeline implementable in Python with HuggingFace Transformers, spaCy, and FastAPI
- Alpaca API free tier covers all historical data needs for backtesting
- Team skill match: Advanced NLP proficiency directly enables fine-tuning and custom architecture work

6.2 Timeline (Pre-Presentation)

- Day 1–2: Data collection (EDGAR scraper + Alpaca price pull)
- Day 3–4: FinBERT fine-tuning + hedge NER model
- Day 5: Signal backtesting + correlation analysis
- Day 6–7: Slide deck + proposal finalization

6.3 Risks & Mitigations

- Risk: Transcript scraping blocked — Mitigation: Fall back to Hugging Face pre-built dataset
- Risk: Low signal correlation — Mitigation: Segment by sector; large-cap tech shows stronger transcript-price link
- Risk: Fine-tuning time — Mitigation: Use ProsusAI/finbert off-the-shelf for demo, fine-tune in background

7. Financial Analysis & Monetization

7.1 Target Market Segments

- Tier 1 — Boutique hedge funds & quant desks (primary): 1,200+ funds in U.S. under \$1B AUM
- Tier 2 — Retail algorithmic traders (secondary): 50,000+ active algo traders needing edge
- Tier 3 — IR departments (future): Companies using sentiment scores for self-monitoring

7.2 Revenue Model

Tier	Price & Model
API Access (Starter)	\$299/mo — 500 calls/month, 3-day delay
Professional	\$999/mo — Unlimited calls, real-time, backtesting dashboard
Enterprise / Hedge Fund	\$4,999/mo — Dedicated fine-tuned model, white-label API, SLA
Data Licensing	\$25,000/yr — Full historical signal dataset (2015–present)

7.3 Unit Economics (Year 1 Conservative Scenario)

- 10 Professional subscribers $\times$ \$999 $\times$ 12 = \$119,880
- 2 Enterprise subscribers $\times$ \$4,999 $\times$ 12 = \$119,976
- 1 Data license = \$25,000
- Estimated Year 1 Revenue: ~\$265,000
- Cloud compute cost (AWS/GCP inference): ~\$18,000/yr
- Gross margin estimate: ~93%

Comparable Valuation Benchmark

SaaS fintech platforms with NLP/alternative data components (e.g., Amenity Analytics, acquired by S&P Global) have sold at 10–20x ARR. At \$265K ARR and a conservative 10x multiple, EarningsEdge implies a ~\$2.65M seed valuation — reasonable for a data-moat business with backtested signal performance.

8. Conclusion

EarningsEdge addresses a well-documented inefficiency at the intersection of NLP and quantitative finance. The technical foundation is achievable with existing open-source models and free data APIs. The monetization path is clear, the target market is well-defined and willing to pay, and the backtestable alpha signal creates a compelling, verifiable value proposition.

The team's advanced NLP capability is a decisive advantage — most competing tools use simple lexicon-based scoring. A fine-tuned FinBERT model with composite hedge-language features represents a meaningful technical moat.

Recommended Next Step

Begin data collection immediately using the SEC EDGAR API and Hugging Face EarningsCall dataset. Run FinBERT off-the-shelf on 50 transcripts as a proof-of-concept before fine-tuning — this gives you presentable results for Tuesday while the full pipeline is in progress.